

Cisco DC Fabrics and Cilium: Deploy, Secure, and Observe Cloud-Native Applications

cisco Live !

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Webex App

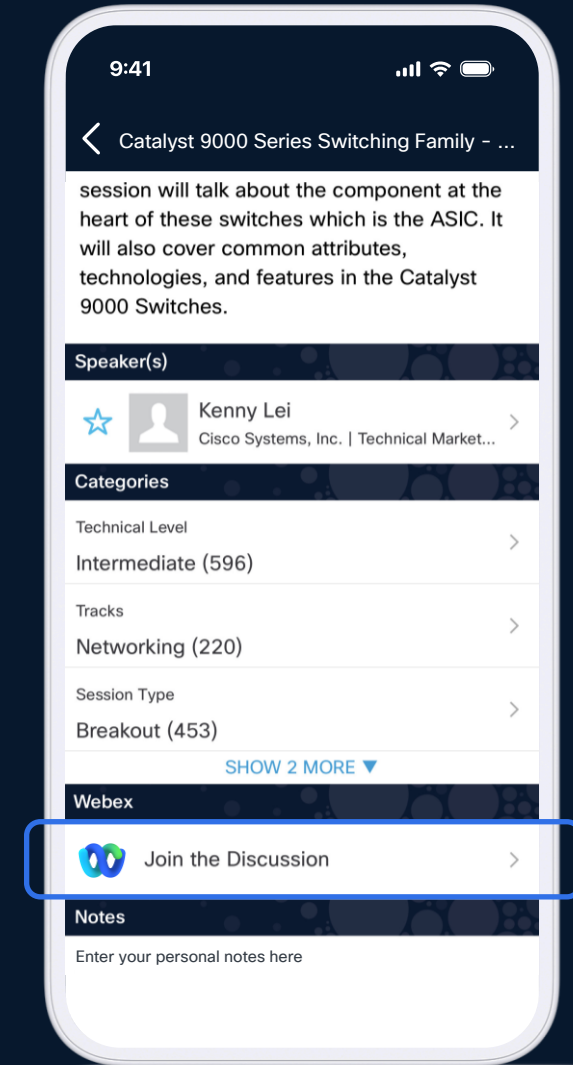
Questions?

Use Webex App to chat with the speaker after the session

How

- 1 Find this session in the Cisco Events App
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Webex spaces will be moderated by the speaker until February 27, 2026.

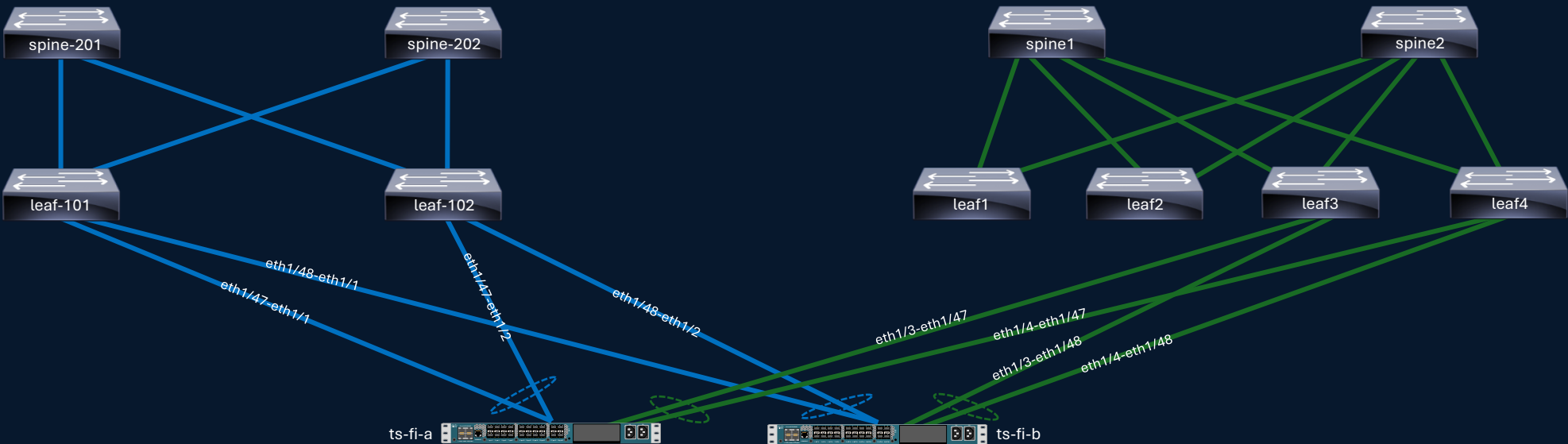


Lab Overview

- Two tracks
- 60 3-node Kubernetes clusters
- 2 Red Hat Openshift clusters for Virtualization and Timescape
- ACI Fabric
- NX-OS EVPN Fabric (managed by ND)

ams-aci (ACI) Routable network 10.61.0.0/16

fabric1 (NX-OS/EVPN) Routable network 10.62.0.0/16



ts-vcenter.amsdmsz.local

- amsterdam
 - titansphere
 - ts-esx1.amsdmsz.local
 - ts-esx2.amsdmsz.local
 - ts-esx3.amsdmsz.local
 - ts-esx4.amsdmsz.local
 - ts-esx5.amsdmsz.local
 - ts-esx6.amsdmsz.local
 - ts-esx7.amsdmsz.local
 - ts-esx8.amsdmsz.local

Disjoint L2 (VLAN 400-600 ACI, VLAN 600-800 NX-OS)

ts-esx* vics	vswitch/dvs	connect ed to fabric
vmnic6	ts-dvs	aci
vmnic7		
vmnic9	ts-dvs-nxos	nxos
vmnic10		

ts-vcenter.amsdmsz.local

- amsterdam
 - vm-network-98
 - ts-dvs
 - aci-vlan-403
 - aci-vlan-404
 - aci-vlan-411
 - aci-vlan-412
 - ts-dvs-uplinks
 - ts-dvs-nxos
 - nxos-vlan-601
 - nxos-vlan-602
 - ts-dvs-nxos-uplinks
 - ts-dvs-nxos-pinned
 - nxos-vlan-601-pinned
 - trunk-nxos
 - ts-dvs-nos-pinned-uplinks
 - ts-dvs-pinned
 - pinned-vlan-403
 - pinned-vlan-404
 - trunk
 - ts-dvs-pinned-uplinks

Virtual Machines

	Name	State
☐	ts-ocp-2-m1	Powered On
☐	ts-ocp-2-m2	Powered On
☐	ts-ocp-2-m3	Powered On
☐	ts-ocp-2-w1	Powered On
☐	ts-ocp-2-w2	Powered On
☐	ts-ocp-2-w3	Powered On
☐	ts-ocp-2-w4	Powered On

What you will learn

1. Kubernetes fundamentals
2. Cilium integration with Cisco ACI & NX-OS VXLAN EVPN
3. Application deployment in Kubernetes
4. Securing and monitoring Application Policies
5. Exposing applications to external clients
6. Securing Connectivity with ACI/NX-OS
7. Egress Gateway and Pod identity Preservation
8. Monitoring Application Health with Nexus Dashboard
9. Deploying Virtual Machines on RedHat OpenShift

What is Cilium?

- Cilium is an open source, cloud native solution for providing, securing, and observing network connectivity between workloads.
- Cilium was created by Isovalent (joined Cisco via acquisition in April 2024.) and is currently the only CNCF graduated networking plugin project.
- The enterprise version is called **Isovalent Networking for Kubernetes**
- **Advanced Network Security:** fine-grained security policies that can be defined based on application-layer identities rather than just IP addresses. This allows for more precise control over communication between microservices.
- **Scalability and Performance:** By using eBPF it achieves high performance and scalability, as it can run network policies directly in the Linux kernel without the need for complex user-space processing.
- **Service Mesh Integration:** integrated with service mesh architectures to provide transparent, high-performance networking and security features that complement service mesh capabilities.
- **Observability:** Robust observability tools, allowing users to monitor and troubleshoot network traffic and security policies with detailed metrics and visibility into application communication patterns.
- **Kubernetes Native:** Designed with Kubernetes in mind seamlessly integrates with Kubernetes to provide native support for network policies and service discovery

eBPF – Foundation of Cilium

- Kubernetes networking
- Load balancing
- Kubernetes services
- Identity-based security
- L7 policies

- Dependencies map (service and flows)
- Monitoring and alerting
- App monitoring

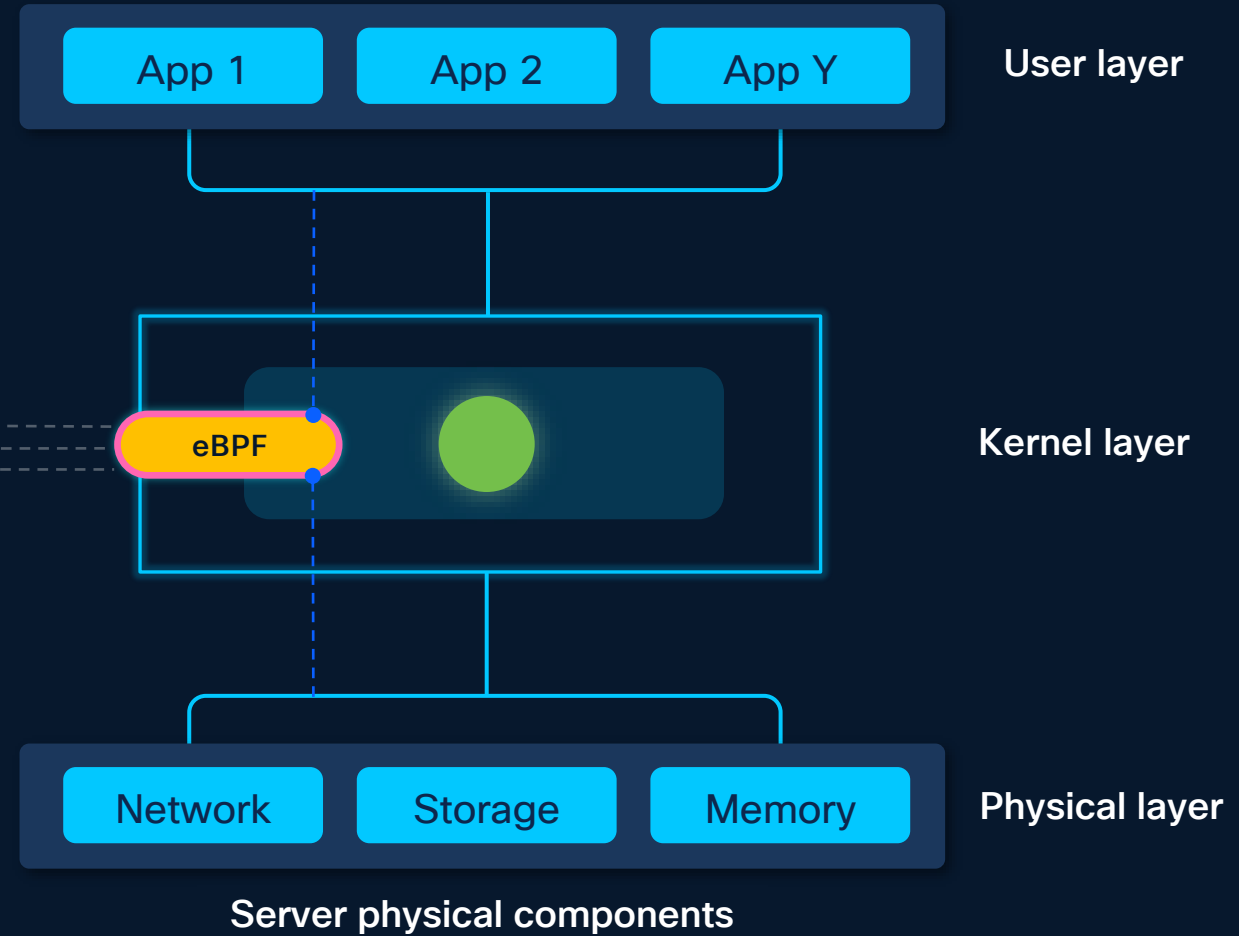
- Monitor process execution
- Runtime security policies
- Real time enforcement



Network filtering

Observability

Security policy



Isovalent and Cisco DC Fabrics – Better Together

- Easy and Scalable:
 - BGP is a well establish protocol easy to configure and troubleshoot
 - High Scale with thousand nodes per cluster
- Visibility:
 - Every communication flow can be monitored with Hubble
 - Traffic Traversing the network can be monitored with conventional tools
- Security:
 - Cloud Native Security enforced by Isovalent using eBPF
 - Network level security (Node to Node, N/S or S/N) leveraging fabric contracts or service insertion
- NetOps Ready:
 - Kubernetes ACI and NX-OS can be driven by API, ansible or terraform ensuring automation is a breeze

How to get started

- Set up a VPN connection with “**vpn.ciscolab.nl**”
- Username “**student-n**” and password “**ins3965!**”
- Open <https://lab-guide.amsdmz.nl>
- When you see password XXXXXX replace it with the password shown above.
- Please remember your student-id.

Design guide available now!



isovalent.com/brief/designing-isovalent-enterprise-for-cisco-aci-and-nexus/

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Book

Guide: Designing Isovalent Enterprise for Cisco ACI & Nexus

Delivering advanced networking, security, and observability for cloud-native workloads on Cisco ACI and Nexus Fabrics.

The Isovalent Enterprise Platform harnesses eBPF-powered technology to integrate seamlessly with Cisco ACI and Nexus 9000 deployments. It provides a robust solution for connecting, securing, and operating dynamic cloud-native workloads within your Cisco infrastructure, ensuring you retain the visibility and control expected from networking architects and practitioners.

Download this Design Guide (40 pages) and learn how the Isovalent Enterprise Platform on Cisco ACI and Nexus fabrics helps you **simplify connectivity** and operations, **strengthen security** with a path to Zero Trust and multi-tenancy control, **maximize performance** for demanding applications including AI workloads, and gain **deep visibility** across your cloud-native environment. Discover how this integration provides network engineers with familiar tooling and ensures robust, scalable infrastructure.

Download the eBook

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